
Litter Recognition 1

Automated Roadside
Litter Collection

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Our Client and Vision

Our client, Ben Scott-Robinson, is the founder of the startup Moss&Gund

The vision of our project is to develop an autonomous robot arm, able to identify and dispose of roadside litter into a bag using custom object detection AI.

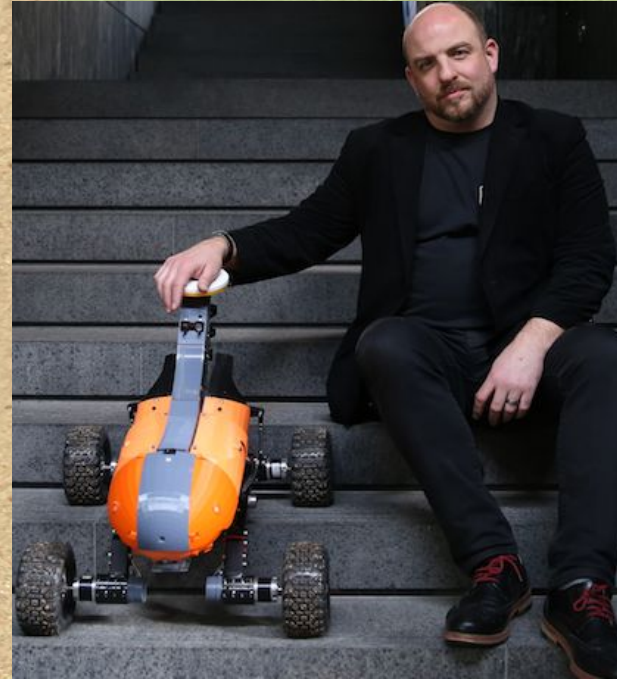
Our first step was to brainstorm user stories to serve as a basis for the requirements and constraints of the project



User Stories

Client

As a client, I want a commercially viable autonomous system to detect and collect litter on the roadside. It should be able to detect at least six key types of litter and be able to sort them accordingly.



User Stories

Waste Management Contractor

As a Waste Management Contractor, I want to be able to integrate autonomous litter collection vehicles into my operations to reduce the cost and increase the safety of roadside litter collection.



Downgrading Client Expectations

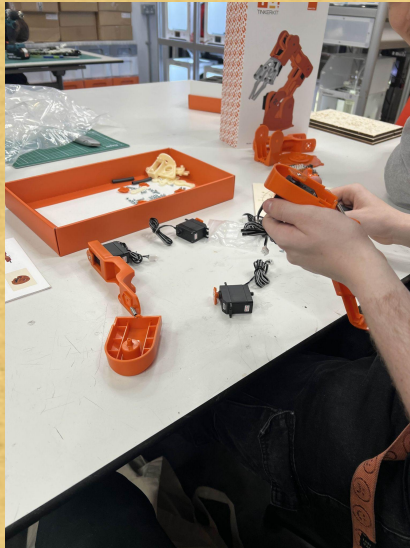
Initially we had been asked to synchronise two robotic arms to complete our task.

We quickly discovered that this was far out of the scope of an SEP project, and so we quickly arranged another meeting with our client and organised our new final goal of just using one arm!



Thank you MVB Hackspace

The biggest hurdle in our project was gaining access to the hardware we'd need. We reached out to the HackSpace and luckily for us they happened to have two, Arduino controlled, Tinkerkit Braccio Robotic Arms that they were more than happy for us to borrow for the year and build!



Into the Unknown of Computer Vision

As a group, we were all novices to computer vision. This has meant we have had to help each other learn and research the topic by creating guides on the topics/technologies we come across.

Research and guidance on how to use tensorflow for object detection

Videos and Guides:

- <https://www.youtube.com/playlist?list=PLAs-3cqyNbjiqaTLHNSu2g4kpaw6TGcUd>
- <https://tensorflow-object-detection-api-tutorial.readthedocs.io/en/latest/install.html>
- https://tensorflow-object-detection-api-tutorial.readthedocs.io/en/latest/auto_examples/object_detection_camera.html#sphx-glr-auto-examples-object-detection-camera-py

Repos used:

- <https://github.com/tensorflow/models> (Some sample models for object detection in TensorFlow)

Tools and Programs used

- miniconda (<https://docs.anaconda.com/miniconda/miniconda-install/>)
- protobuf (<https://github.com/protocolbuffers/protobuf/releases>)

Installation Guide

Install miniconda

Go to <https://docs.anaconda.com/miniconda/miniconda-install/> and install the right version for your machine. Run the following in a new

PyTorch Labsheets

Labsheets for the Applied Deep Learning course.

Overview

Labsheet	Description
<u>0</u>	Introduction to Python and the scientific Python ecosystem
<u>1</u>	Your First Fully Connected Network
<u>2</u>	Your First CNN
<u>3</u>	Techniques for Training DNNs
<u>4</u>	Data Augmentation
<u>5</u>	Transformers

Development Testing

```
name: Python application

on:
  push:
    branches: [ "dev" ]
  pull_request:
    branches: [ "dev" ]

permissions:
  contents: read

jobs:
  build:

    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4
      - name: Set up Python 3.10
        uses: actions/setup-python@v5
        with:
          python-version: '3.x'
      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install -r code/requirements.txt
      - name: Test with pytest
        run: |
          pip install pytest pytest-cov
          pytest code/ --doctest-modules --junitxml=junit/test-results.xml --cov=com --cov-report=xml --cov-report=html
```

Throughout the development of our project, we have been using a Python continuous integration action to ensure our code is thoroughly tested.

Furthermore given the manner of our project, we have been able to test a lot of our code in person as is shown on our project demonstration video on the next slide.



```
python3 keylogger.py
Press the key to see which key is pressed. Press ESC to exit.
Key History (last 5):
1: q
2: w
3: e
4: r
5: t
Last Key Pressed: d
```



Project Management

Stand Ups with Client

Our client has been heavily involved in our development cycle, with us choosing to meet every 2 weeks.

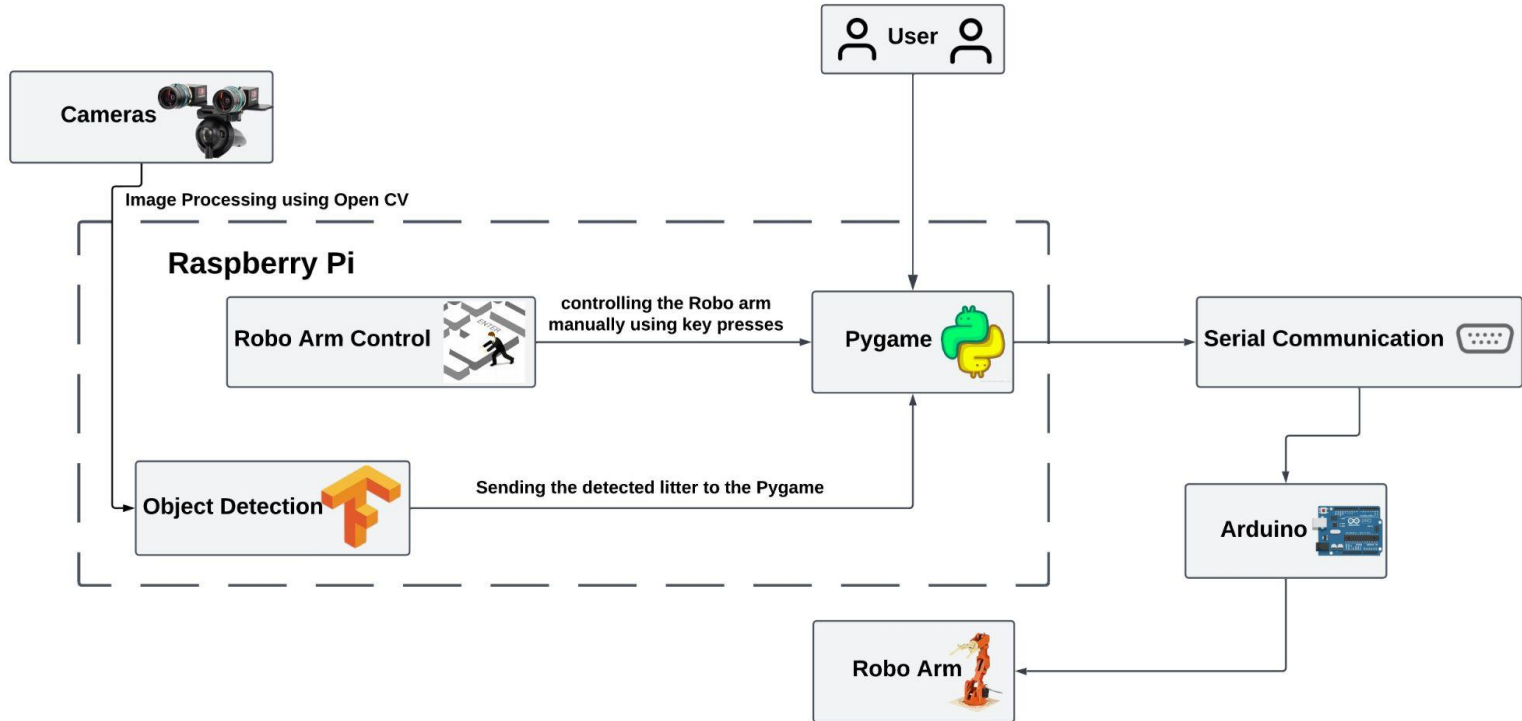
Weekly Hackspace Meetings

Each Wednesday morning, we have been meeting in MVB to test our weekly progress on our robotic arm in the HackSpace.

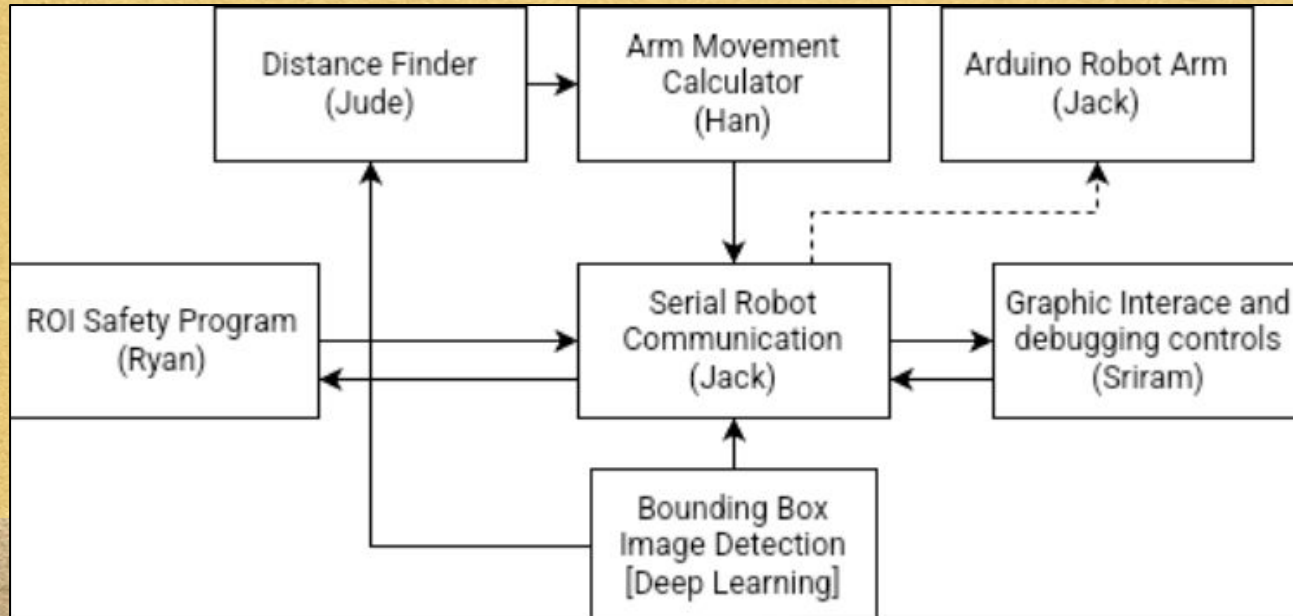
Github Version Control

Using the version control capabilities of GitHub, we have been able to split up our code workflow between members of our group.

Architecture Diagram



How has the Project has been SEParated?



Client Feedback

- Use RetinaNet
- Move to anywhere
- Integration



What Comes Next?

- Do the research
- Distance calculator and Degree calculator
- Integration

Questions